



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

SQL Fundamental for Beginners

TGS Ref No: TGS-2020505790

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 14

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
11.0	16 Oct 2020	Legacy lesson plan accompanying the v11 slide deck.	Dr. Alfred Ang
12.0	8 August 2026	Full rebuild in the single-source pipeline: visual slide deck (no in-deck step-by-step), 11 hands-on labs across 4 topics, PP + OQ assessment per Assessment Plan v1.2, schedule with slide references.	Dr. Alfred Ang
13.0	11 August 2026	Labs re-based on the SG Mart realistic mock dataset supplied per lab as CSV, Excel and SQL seed scripts; lab timings unchanged; all lab SQL and expected results verified against the shipped database.	Dr. Alfred Ang
14	11 August 2026	Each lab is now one self-contained folder holding its instructions, CSV data and SQL seed scripts together; schedule and timings unchanged.	Dr. Alfred Ang

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Course Information

Course Title	SQL Fundamental for Beginners
WSQ Course Reference	TGS-2020505790
Skills Framework	Data Engineering (ICT-DIT-3005-1.1), ICT Skills Framework
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 8 training hours
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on SQL labs in SQLite Studio on the SG Mart sample database (sgmart.db)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply data modeling for business processes.
- LO2: Apply data processing and analysis using SQL.
- LO3: Apply data transformation from multiple data sources.
- LO4: Apply data mapping to data warehouse.

Assessment

- Practical Performance (PP) — hands-on SQL tasks, 70 minutes, open book, assessor-to-candidate ratio 1:10.
- Oral Questioning (OQ) — 20 minutes, one-to-one with the assessor.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted at the end of the day: Briefing 4:45 pm, PP 5:00–6:10 pm, OQ 6:10–6:30 pm.
- A minimum of 75% attendance is required and learners must be assessed as 'Competent' to be eligible for funding.

Course Schedule

Day 1 — Data Modeling, SQL Queries, Transformation & Data Warehouse

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, trainer and learner introductions, ground rules, downloading course material, the SG Mart course dataset, lesson plan, Skills Framework, learning outcomes and mandatory digital attendance

		(AM). Slides 1–25
10:00–11:00	60 min	SQL Foundations: what SQL is and can do, history, SQL databases, tables, SQLite Studio. Topic 1 — Data Modeling: CREATE DATABASE/TABLE, data types, constraints, keys and data modeling (concepts + demo). Hands-on: Lab 1: Set Up SQLite Studio & Import the SG Mart Database; Lab 2: Create a Database and Tables. Slides 26–54
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	Hands-on: Lab 3: Model Data with Constraints and Keys. Topic 2 — Data Processing and Analysis: SELECT, DISTINCT, WHERE, operators, ORDER BY, LIMIT, aliases (concepts + demo). Hands-on: Lab 4: Query Data with SELECT. Slides 55–66
12:30–13:30	60 min	INSERT, UPDATE, DELETE (concepts + demo). Hands-on: Lab 5: Filter with SQL Operators; Lab 6: Insert, Update and Delete Records. Slides 67–81
13:30–14:30	60 min	Lunch break
14:30–15:30	60 min	Topic 3 — Data Transformation: aggregate functions, joins, UNION, GROUP BY, HAVING (concepts + demo). Hands-on: Lab 7: Aggregate Data with COUNT, AVG and SUM; Lab 8: Join Data from Multiple Tables; Lab 9: Group Data with GROUP BY and HAVING. Slides 82–103
15:30–15:45	15 min	Tea break
15:45–16:30	45 min	Topic 4 — Introduction to Data Warehouse: warehouse concepts, OLAP, RDBMS, E-R model, SMRT case study, stored procedures (concepts + demo). Hands-on: Lab 10: Map an E-R Model to Database Tables (SMRT Case Study); Lab 11: Author Stored Procedures for SQL Server. Slides 104–123
16:30–16:45	15 min	Course recap and summary, Q&A, recommended courses, TRAQOM course-feedback survey and Assessment digital

		attendance. Slide 124
16:45–17:00	15 min	Briefing for Assessment. Slide 16
17:00–18:10	70 min	Final Assessment — Practical Performance (PP): hands-on SQL tasks, 70 minutes, open book
18:10–18:30	20 min	Final Assessment — Oral Questioning (OQ): 20 minutes, one-to-one. Sign the Assessment Summary Record. Slides 129–132

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the TSC)

Topic	TSC Mapping	Labs
Topic 01: Data Modeling	A1, A2, K1, K2	Lab 1: Set Up SQLite Studio & Import the SG Mart Database; Lab 2: Create a Database and Tables; Lab 3: Model Data with Constraints and Keys
Topic 02: Data Processing and Analysis	A3, A5, K4	Lab 4: Query Data with SELECT; Lab 5: Filter with SQL Operators; Lab 6: Insert, Update and Delete Records
Topic 03: Data Transformation	A4, A6, K5, K6	Lab 7: Aggregate Data with COUNT, AVG and SUM; Lab 8: Join Data from Multiple Tables; Lab 9: Group Data with GROUP BY and HAVING
Topic 04: Introduction to Data Warehouse	A7, A8, A9, K3, K7	Lab 10: Map an E-R Model to Database Tables (SMRT Case Study); Lab 11: Author Stored Procedures for SQL Server